Microsoft Intune Zero-to-Hero

COMPLETE PRACTICAL DEVICE MANAGEMENT & SECURITY MASTERCLASS

SECTION 1 — Intune Fundamentals & Tenant Setup

"BUILD THE FOUNDATION FOR A SECURE, MODERN INTUNE ENVIRONMENT."

TRAINING INTRODUCTION

- **Why Intune Feels Hard in Practice**
  - Policies show Successful but don't apply
  - Apps fail without clear reasons
  - Compliance behaves unexpectedly
  - Conditional Access confusion

- **Not Theory. Real Enterprise Experience**
  - No marketing demos
  - No certification-only slides
  - Real production scenarios
  - Endpoint-first troubleshooting

- **Skills You Will Gain**
  - Why Intune settings exist
  - How policies are evaluated
  - How endpoints respond
  - How to validate & troubleshoot
  - How to explain decisions clearly

- **36 Real Enterprise Labs**
  - Windows 10 & Windows 11
  - iOS & Android enrollment concepts
  - Compliance & configuration
  - App deployment & detection rules
  - Endpoint Security & ASR

SECTION 1 — Intune Fundamentals & Tenant Setup

"BUILD THE FOUNDATION FOR A SECURE, MODERN INTUNE ENVIRONMENT."

TRAINING INTRODUCTION

- **Is this course Right for you?**
  - IT Administrators
  - Endpoint & System Engineers
  - Security Analysts
  - Students entering Intune roles
- **From Guessing to Confidence**
  - Deploy with confidence
  - Secure with clarity
  - Troubleshoot with logic
  - Own the platform

SECTION 1: Intune Fundamentals & Tenant Setup

"BUILD THE FOUNDATION FOR A SECURE, MODERN INTUNE ENVIRONMENT."

SECTION 1 — Intune Fundamentals & Tenant Setup

"BUILD THE FOUNDATION FOR A SECURE, MODERN INTUNE ENVIRONMENT."

WHAT YOU WILL LEARN IN SECTION 1

- **1.1. What Is Microsoft Intune (MDM/MAM Overview)**
  - Learn the purpose of Intune
  - Understand MDM vs MAM
  - See how Intune manages corporate and BYOD devices
  - Review the full Intune device → policy → compliance → access flow using the diagram
- **1.2. Intune vs Configuration Manager (SCCM)**
  - Understand the difference between cloud-based Intune and on-prem SCCM
  - Learn why organizations are shifting to modern cloud management
  - See how hybrid coexistence works under Microsoft Endpoint Manager
- **1.3. Unified Endpoint Security: Intune + Entra ID + Defender**
  - Learn how identity, device posture, and threat protection work together
  - See how Conditional Access evaluates identity + device + risk
  - Understand why this forms Microsoft's Zero Trust endpoint architecture

SECTION 1 — Intune Fundamentals & Tenant Setup

"BUILD THE FOUNDATION FOR A SECURE, MODERN INTUNE ENVIRONMENT."

WHAT YOU WILL LEARN IN SECTION 1

- **1.4. Licensing Requirements (E5 Capabilities)**
  - Understand what Intune features are unlocked in Microsoft 365 E5
  - Learn why your training environment has full access
  - See which security features we'll use in upcoming labs
- **1.5. Supported Device Platforms**
  - Discover which OS platforms Intune can manage
  - Understand how Windows, macOS, iOS/iPadOS, Android, and even Linux fit into modern endpoint management
  - Learn why Windows is the primary platform for this course
- **1.6. Transition to Labs & Environment Preparation**
  - Review the core tasks required before device enrollment
  - Understand why licensing, MDM/MAM authority, and user setup must happen first
  - Prepare for three foundational labs that configure your tenant for real-world use

SECTION 1 — Intune Fundamentals & Tenant Setup

“BUILD THE FOUNDATION FOR A SECURE, MODERN INTUNE ENVIRONMENT.”

WHAT YOU WILL LEARN IN SECTION 1

- **AFTER THIS THEORY, YOU WILL COMPLETE THREE LABS:**
  - Lab 1.1 — Verify Intune Licensing & Portal Access
  - Lab 1.2 — Configure MDM + MAM Authority
  - Lab 1.3 — Create Intune Admin & Test User Accounts

SECTION 1 — Intune Fundamentals & Tenant Setup

“BUILD THE FOUNDATION FOR A SECURE, MODERN INTUNE ENVIRONMENT.”

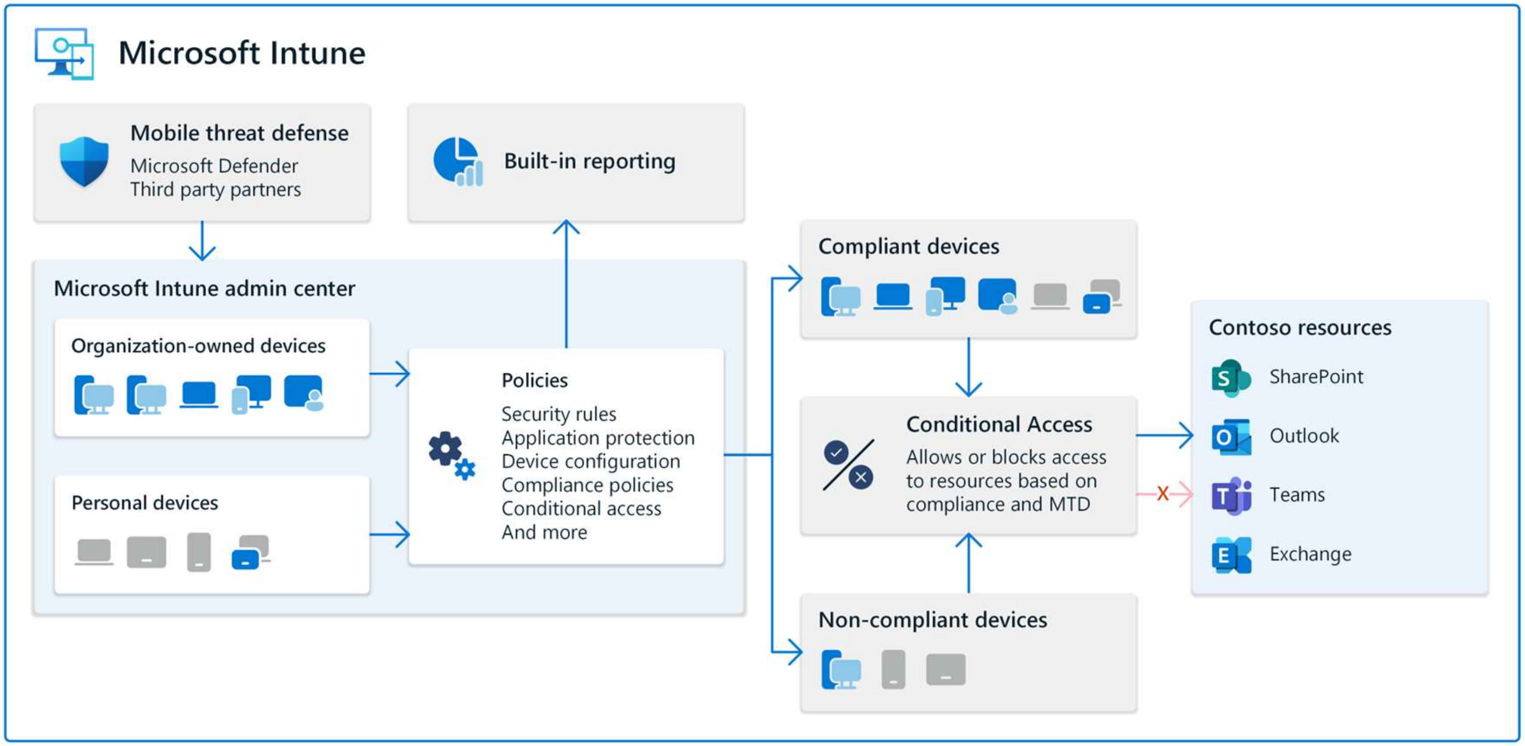
1.1 WHAT IS INTUNE (MDM/MAM OVERVIEW)

- **MDM — Mobile Device Management**
  - Full device-level control
  - Enforce security settings (passwords, encryption, firewall, AV)
  - Configure Wi-Fi, VPN, certificates, restrictions
  - Validate device compliance before access
- **MAM — Mobile Application Management**
  - Protects corporate data inside apps, not the device
  - Perfect for BYOD / personal devices
  - App-level controls (PIN, encryption, block copy/paste)
  - Selective wipe removes only corporate data
  - Applies to Outlook, Teams, Edge & Microsoft 365 apps
- **MDM vs. MAM — Key Difference**
  - MDM = Manage the entire device
  - MAM = Manage corporate data inside applications
  - Used together for secure, Zero Trust access
  - Core concept for all future Intune configuration decisions

SECTION 1 — Intune Fundamentals & Tenant Setup

"BUILD THE FOUNDATION FOR A SECURE, MODERN INTUNE ENVIRONMENT."

1.1 WHAT IS INTUNE (MDM/MAM OVERVIEW)



SECTION 1 — Intune Fundamentals & Tenant Setup

"BUILD THE FOUNDATION FOR A SECURE, MODERN INTUNE ENVIRONMENT."

1.2 INTUNE VS. CONFIGURATION (SCCM)

SCCM — Traditional On-Premises Management

- Mature, powerful platform for Windows environments
- Designed for devices inside corporate networks
- Requires on-prem servers & distribution points
- Depends on internal infrastructure & sometimes VPN
- Ideal for legacy or perimeter-based management models

Intune — Modern Cloud-Based Management

- 100% cloud-native; no servers required
- Microsoft manages all backend infrastructure
- Works with Windows, macOS, iOS, Android
- Manages devices anywhere with internet access
- Ideal for remote work, contractors, and global teams

Modern Needs vs Legacy Needs

- Intune supports hybrid, remote, and distributed workforces
- Reduces infrastructure cost & operational complexity
- Built for Zero Trust: no reliance on corporate perimeter
- SCCM still required for some legacy or specialized apps
- Microsoft supports hybrid management: Intune + SCCM (Endpoint Manager)

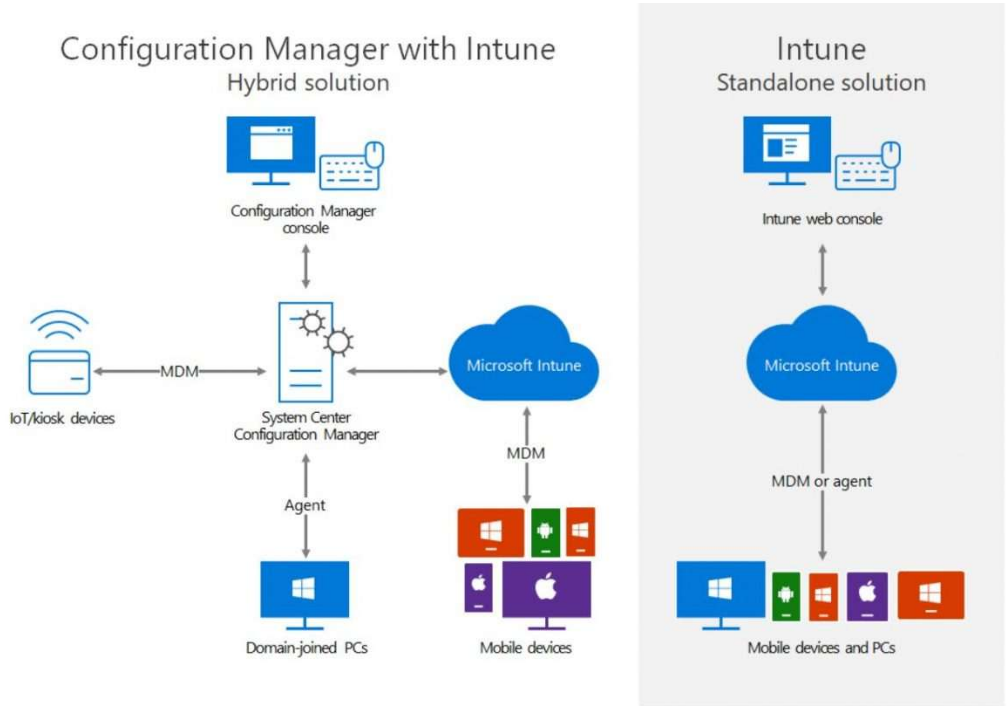
The Future of Device Management

- Industry trend is moving toward cloud management
- Intune is becoming the default tool for most devices
- SCCM remains relevant for specific, specialized scenarios
- Long-term strategy: cloud-first, modern endpoint management

SECTION 1 — Intune Fundamentals & Tenant Setup

"BUILD THE FOUNDATION FOR A SECURE, MODERN INTUNE ENVIRONMENT."

1.2 INTUNE VS. CONFIGURATION (SCCM)



SECTION 1 — Intune Fundamentals & Tenant Setup

"BUILD THE FOUNDATION FOR A SECURE, MODERN INTUNE ENVIRONMENT."

1.3 INTUNE + ENTRA ID + DEFENDER = UNIFIED ENDPOINT SECURITY

Entra ID — Identity Verification

- Verifies who the user is
- Performs identity risk analysis & MFA checks
- Applies Conditional Access rules
- Ensures access only for trusted, authenticated users
- Asks: **Is this the right user**?

Intune — Device Compliance & Health

- Verifies device health and security posture
- Checks encryption, OS updates, antivirus & firewall
- Evaluates ASR rules & other compliance signals
- Enforces remediation or blocks non-compliant devices
- Asks: **Is this device trusted**?

Microsoft Defender — Threat Detection & Risk

- Real-time detection of attacks & suspicious activity
- Uses MITRE ATT&CK–based behavior analytics
- Performs automated investigations & response
- Can isolate compromised devices
- Asks: **Is this device safe right now**?

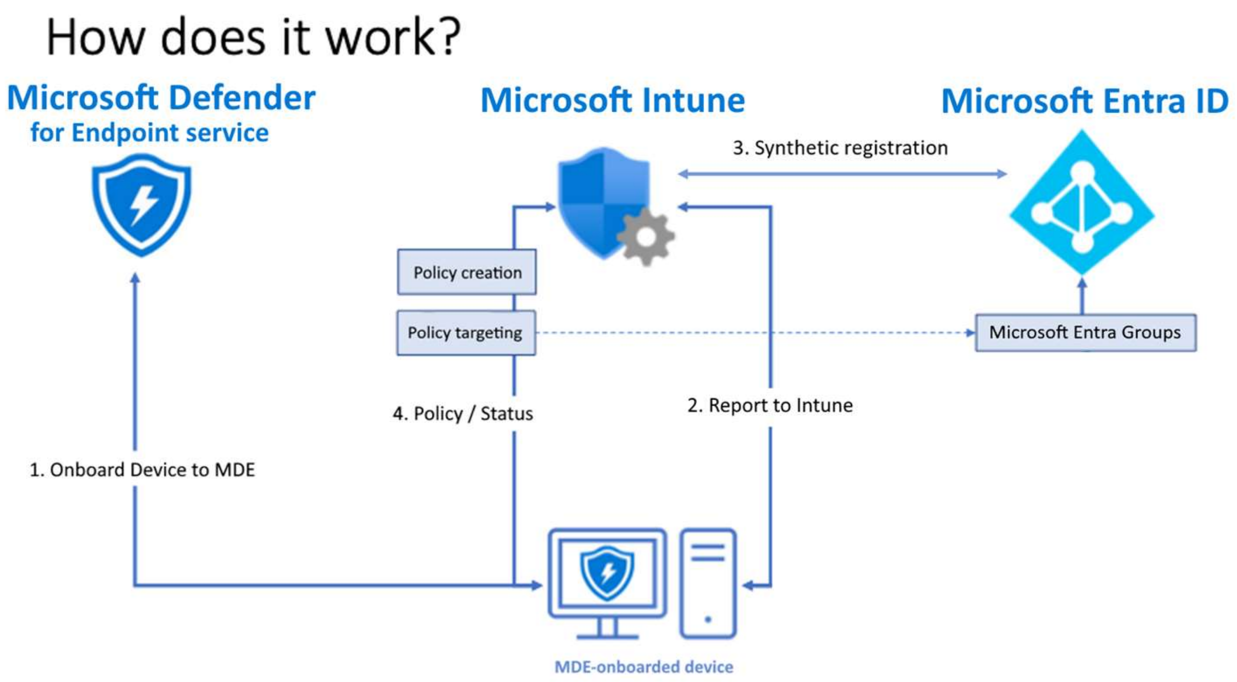
Zero Trust Unified Access Decision

- **Entra ID**: Is the user trusted?
- **Intune**: Is the device trusted?
- **Defender**: Is the device safe?
- Together form unified endpoint security
- Enables continuous, real-time protection & access control

SECTION 1 — Intune Fundamentals & Tenant Setup

"BUILD THE FOUNDATION FOR A SECURE, MODERN INTUNE ENVIRONMENT."

1.3 INTUNE + ENTRA ID + DEFENDER = UNIFIED ENDPOINT SECURITY



SECTION 1 — Intune Fundamentals & Tenant Setup

"BUILD THE FOUNDATION FOR A SECURE, MODERN INTUNE ENVIRONMENT."

1.4 LICENSING REQUIREMENTS

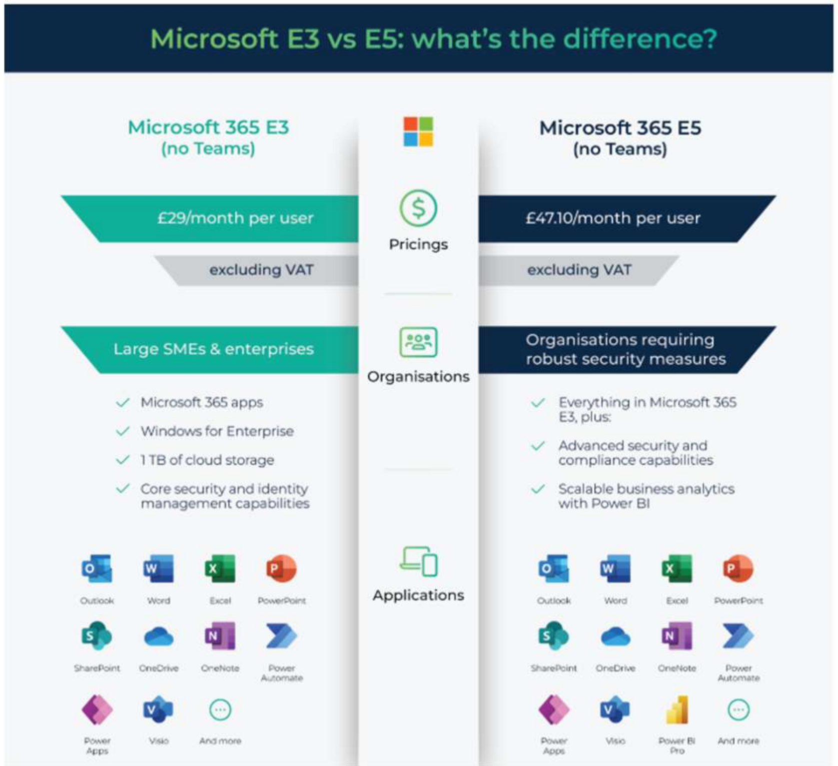
- **What E5 Includes**
  - Full MDM: Configure & secure entire devices
  - Full MAM: Protect corporate data inside apps
  - Built-in Microsoft Security Baselines
  - Compliance Policies to enforce device posture
  - Conditional Access for Zero Trust evaluation
  - Microsoft Defender for Endpoint, including:
  - Device risk scoring & threat analytics
  - Attack timelines & behavioral detection
  - Automated investigation & remediation
  - Device isolation for compromised endpoints
  - Attack Surface Reduction (ASR) rules
  - Vulnerability management & advanced reporting
  - Deep integration across Microsoft's security ecosystem

- **Why E5 Matters for This Course**
  - Every lab and policy is fully supported
  - No restricted or unavailable features
  - You are working with enterprise-grade capabilities
  - Training reflects real-world, production environments

SECTION 1 — Intune Fundamentals & Tenant Setup

"BUILD THE FOUNDATION FOR A SECURE, MODERN INTUNE ENVIRONMENT."

1.4 LICENSING REQUIREMENTS



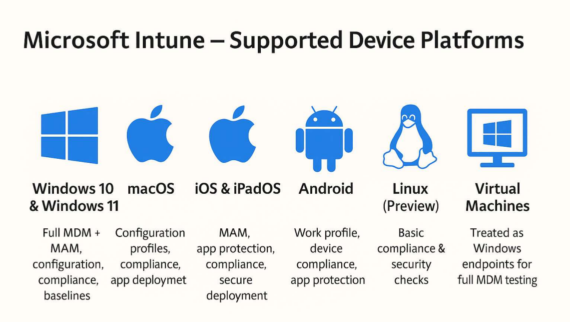
SECTION 1 — Intune Fundamentals & Tenant Setup

"BUILD THE FOUNDATION FOR A SECURE, MODERN INTUNE ENVIRONMENT."

1.5 SUPPORTED DEVICE TYPES

- **Supported Platforms in Intune**
  - **Windows 10 & Windows 11:**
    - Most complete management capabilities
    - Deep configuration, compliance & security control
  - **macOS:**
    - Profiles, compliance, security, and app deployment
    - Valuable for mixed Apple + Windows environments
  - **iOS / iPadOS:**
    - Strong app protection & compliance enforcement
  - **Android:**
    - Supports Work Profile, Corporate-Owned, and Fully Managed modes
  - **Linux (Preview):**
    - Basic compliance & security posture controls
  - **Virtual Machines (VMs):**
    - Treated the same as physical devices
    - Ideal for lab and training scenarios

- **Why Multi-Platform Support Matters**
  - Real-world organizations rarely use a single platform
  - Intune enables consistent security & compliance across all devices
  - Centralized management for desktop, mobile & virtual endpoints
  - This course focuses on Windows for the richest hands-on experience



SECTION 1 — Intune Fundamentals & Tenant Setup

"BUILD THE FOUNDATION FOR A SECURE, MODERN INTUNE ENVIRONMENT."

SECTION OVERVIEW & TRANSITION TO LABS

- **Your Three Essential Setup Labs**
  1. **Lab 1.1 — Verify Licensing & Portal Access**
     1. Confirm Intune + E5 components are active
     2. Ensure access to the Intune Admin Center
  2. **Lab 1.2 — Configure MDM & MAM Authority**
     1. Define the tenant's device management authority
     2. Select which users can enroll devices
  3. **Lab 1.3 — Create Required User Accounts**
     1. Dedicated Intune Admin account
     2. Test User account for device enrollment & scenario testing

- **What Happens After These Labs?**
  - Tenant fully prepared for device enrollment
  - Ready for Section 2: Windows 10/11 enrollment
  - Begin validating Intune management and compliance
  - Next step: **Lab 1.1 — Verify Licensing & Portal Access**

Lab 1.1: Verify Intune Licensing & Portal Access

STEP-BY-STEP INSTRUCTIONS

Lab 1.2: Configure MDM + WIP Authority

STEP-BY-STEP INSTRUCTIONS

Lab 1.3: Create Intune Admin + Test User

STEP-BY-STEP INSTRUCTIONS

SECTION 1 — Intune Fundamentals & Tenant Setup

"BUILD THE FOUNDATION FOR A SECURE, MODERN INTUNE ENVIRONMENT."

LET'S WRAP-UP SECTION 1:-

- **Your Three Essential Setup Labs**
- **Intune is the foundation of modern device management**
- **MDM and MAM serve different but complementary purposes**
- **Intune works with Entra ID and Defender as one ecosystem**
- **Licensing enables capabilities — configuration enables control**
- **A correctly prepared tenant is critical for everything that follows**

SECTION 2 A: Windows Enrollment Deep Dive

“TURN WINDOWS DEVICES INTO FULLY MANAGED, CLOUD-CONTROLLED ENDPOINTS.”

SECTION 2 — Windows Enrollment Deep Dive

"TURN WINDOWS DEVICES INTO FULLY MANAGED, CLOUD-CONTROLLED ENDPOINTS."

WHAT YOU WILL LEARN:-

2.1. Why enrollment method matters

- High-level comparison:
  - Entra ID Join
  - Hybrid Entra ID Join
  - Workgroup → MDM enrollment
- Which methods are supported in modern cloud-first environments

2.2. Entra ID Join + Intune Auto-Enrollment

- What Entra ID Join actually does
- Relationship between:
  - Entra ID
  - MDM auto-enrollment
  - User license + MDM scope
- Why auto-enrollment fails when prerequisites are missing

2.3. What "Successful Enrollment" Really Means

- Difference between:
  - Device appearing in Entra ID
  - Device appearing in Intune
- What Intune checks after enrollment:
  - Ownership
  - MDM authority
  - Primary user
  - Compliance readiness
- Common enrollment misconceptions

SECTION 2 — Windows Enrollment Deep Dive

"TURN WINDOWS DEVICES INTO FULLY MANAGED, CLOUD-CONTROLLED ENDPOINTS."

2.1. WHY ENROLLMENT METHOD MATTERS

Windows Enrollment Methods

- Why enrollment method matters
- Impacts management, security, and troubleshooting
- Foundation for all Intune labs

Entra ID Join (Cloud-First Approach)

- Device joins directly to Entra ID
- Automatic Intune enrollment
- No on-prem infrastructure required
- Ideal for remote and hybrid work

Hybrid Entra ID Join

- Domain-joined to on-prem Active Directory
- Also registered in Entra ID
- Supports legacy apps and Group Policy
- More complex configuration

Workgroup / MDM-Only Enrollment

- Device not joined to Entra ID
- Enrolled using work account
- Common for BYOD and contractors
- Limited management compared to join

Enrollment Methods Comparison

- Entra ID Join → Modern, cloud-first, recommended
- Hybrid Join → Legacy + cloud coexistence
- MDM-Only → Personal or limited-use devices

Enrollment Method Used in This Course

- Entra ID Join
- Automatic Intune enrollment
- Cloud-first design
- Full Intune capabilities

SECTION 2 — Windows Enrollment Deep Dive

"TURN WINDOWS DEVICES INTO FULLY MANAGED, CLOUD-CONTROLLED ENDPOINTS."

2.2. ENTRA ID JOIN + INTUNE AUTO-ENROLLMENT

- **Entra ID Join & Automatic Intune Enrollment**
  - Foundation of cloud-first device management
  - Identity + device management working together
  - No manual enrollment required
- **What Happens During Entra ID Join**
  - Device registers directly in Entra ID
  - Device becomes a cloud identity
  - No on-prem Active Directory required
  - Enables Conditional Access & compliance
- **How Intune Auto-Enrollment Is Triggered**
  - Happens automatically after Entra ID Join
  - No manual enrollment steps
  - Depends on tenant configuration
- **Auto-Enrollment Prerequisites**
  - Entra ID recognizes user & device
  - User has Intune-eligible license
  - User is included in MDM user scope
- **Entra ID ↔ Intune Enrollment Relationship**
  - Entra ID → authentication & device registration
  - Intune → device management
  - MDM user scope connects the two
- **Common Auto-Enrollment Failures**
  - No Intune license assigned
  - User excluded from MDM scope
  - MDM authority not set
  - Device joined before tenant setup
  - Entra ID Join without Intune enabled

SECTION 2 — Windows Enrollment Deep Dive

“TURN WINDOWS DEVICES INTO FULLY MANAGED, CLOUD-CONTROLLED ENDPOINTS.”

2.2. ENTRA ID JOIN + INTUNE AUTO-ENROLLMENT

- **Entra ID Device ≠ Intune-Managed Device**
  - Device may appear in Entra ID
  - Does NOT guarantee Intune management
  - Auto-enrollment must succeed
- **Successful Entra ID Join + Auto-Enrollment**
  - User signs in with work account
  - Device joins Entra ID
  - Licensing & MDM scope validated
  - Intune enrolls device automatically
- **Next: See This in Action**
  - Entra ID Join using Test User
  - Automatic Intune enrollment
  - Real device, real validation

SECTION 2 — Windows Enrollment Deep Dive

“TURN WINDOWS DEVICES INTO FULLY MANAGED, CLOUD-CONTROLLED ENDPOINTS.”

2.3. WHAT “SUCCESSFUL ENROLLMENT” REALLY MEANS

- **What Does “Successfully Enrolled” Really Mean?**
  - Enrollment ≠ just device visibility
  - Identity vs device management
  - Common Intune misunderstanding
- **Registration ≠ Management**
  - Entra ID → device registration
  - Intune → device management
  - Different platforms, different roles
- **When a Device Appears in Entra ID**
  - Device identity exists in the cloud
  - Used for authentication & access
  - Does NOT guarantee Intune management
- **When a Device Appears in Intune**
  - Device is enrolled for management
  - Policies can be deployed
  - Compliance can be evaluated
- **What Intune Checks After Enrollment**
  - Device ownership
  - MDM authority
  - Primary user assignment
  - Compliance readiness
- **Device Ownership & MDM Authority Matter**
  - Corporate vs Personal device
  - Determines policy scope
  - Confirms Intune is the active MDM

SECTION 2 — Windows Enrollment Deep Dive

“TURN WINDOWS DEVICES INTO FULLY MANAGED, CLOUD-CONTROLLED ENDPOINTS.”

2.3. WHAT “SUCCESSFUL ENROLLMENT” REALLY MEANS

- **Primary User & Compliance Status**
  - Primary user drives policy targeting
  - Compliance evaluated after enrollment
  - Initial non-compliance is normal
- **Common Enrollment Misconceptions**
  - Entra ID device = Intune managed ❌
  - Enrolled device = compliant ❌
  - Policies apply instantly ❌
- **Signs of Successful Enrollment**
  - Appears in Intune
  - Intune listed as MDM authority
  - Primary user assigned
  - Policies & compliance applied
- **Next: Validate Enrollment in Real Labs**
  - Verify Intune enrollment
  - Confirm management state
  - Prepare for policies & security

Lab 2.1A: Prepare Windows 10/11 Device for Intune Enrollment

STEP-BY-STEP INSTRUCTIONS

Lab 2.2A: Entra ID Join & Automatic Intune Enrollment

STEP-BY-STEP INSTRUCTIONS

Lab 2.3A: Validate Enrollment & MDM Management State

STEP-BY-STEP INSTRUCTIONS

SECTION 2 B: Mobile Device Enrollment (Android & iOS)

"BRING EVERY MOBILE DEVICE UNDER CONTROL"

SECTION 2B — Mobile Device Enrollment (Android & iOS)

"BRING EVERY MOBILE DEVICE UNDER CONTROL"

WHAT YOU WILL LEARN IN SECTION 2B:-

1.1B. Prepare Intune for Mobile Device Enrollment

- Enable Android Enterprise using Managed Google Play
- Configure required platform prerequisites for Android and iOS
- Understand why mobile platforms require extra trust integrations

1.2B. Enroll Android BYOD Devices Using Work Profile

- Understand Android enrollment models (BYOD vs Corporate)
- Enroll a personal Android device using Company Portal
- See how Work Profile separates work and personal data
- Validate Android device ownership and management type in Intune

1.3B. Configure Apple APNs for iOS Management

- Understand why Apple requires APNs for device management
- Create and upload the Apple MDM Push Certificate
- Enable Intune to securely communicate with iOS/iPadOS devices

1.4B. Enroll iOS Devices Using Company Portal

- Enroll an iPhone or iPad using Company Portal
- Install and approve the MDM management profile
- Verify device status, compliance, and management in Intune

1.5B. Validate and Compare Mobile Device Management

- Confirm Android and iOS devices appear correctly in Intune
- Review device ownership, platform, and management state
- Understand key differences between mobile MDM and Windows MDM

SECTION 2B — Mobile Device Enrollment (Android & iOS)

"BRING EVERY MOBILE DEVICE UNDER CONTROL"

1.1B. PREPARE INTUNE FOR MOBILE DEVICE ENROLLMENT

- **Mobile platforms require platform-specific trust**
  - Contrast with Windows ("Microsoft controls the stack end-to-end")
  - Android and iOS owned by Google/Apple
  - Microsoft must be explicitly trusted

  This directly reinforces why platform-specific trust exists.

- **Android requires Managed Google Play integration**
  - Names Managed Google Play
  - Explains Android Enterprise
  - Lists what breaks without it (Work Profile, apps, compliance)
  - Emphasizes "enrollment fails completely"

  This goes beyond the bullet in a good way.

- **iOS requires Apple Push Notification Service (APNs)**
  - Clearly explains APNs purpose
  - Explains what APNs allows (commands, not data)
  - Covers impact of certificate expiry

  This perfectly matches the slide intent.

- **These are tenant-level prerequisites**
  - "Tenant-level prerequisites"
  - Not per device, not per user
  - One mistake affects all devices

  This is direct alignment, not inferred.

- **Without them, enrollment will fail**
  - "Enrollment doesn't partially work — it fails entirely"
  - "Management stops immediately"
  - "Everything that follows breaks"

  The script emphasizes this more strongly than the slide — which is good teaching.

SECTION 2B — Mobile Device Enrollment (Android & iOS)

"BRING EVERY MOBILE DEVICE UNDER CONTROL"

1.2B. ENROLL ANDROID BYOD DEVICES USING WORK PROFILE

- **Android Enterprise supports multiple enrollment models**
  - Mentions corporate-owned, shared, kiosk scenarios
  - Clearly frames Work Profile as one of several models
- **Most common model: Personally-Owned with Work Profile**
  - "Most common model by far"
  - Directly framed as the standard BYOD scenario
- **Work Profile creates a secure corporate container**
  - Described as a secure, encrypted container
  - "Phone inside a phone" analogy reinforces understanding
- **Work and personal data are kept separate**
  - Clear separation of work apps vs personal data
  - Explicitly states what IT cannot see or control
- **Enrollment performed using Company Portal**
  - Mentions Company Portal installation and sign-in
  - Work Profile creation triggered during enrollment
- **Device validated in Intune (ownership, platform, management type)**
  - Calls out ownership = Personal
  - Platform = Android
  - Management type = Android Enterprise
  - Emphasizes why validation matters

SECTION 2B — Mobile Device Enrollment (Android & iOS)

"BRING EVERY MOBILE DEVICE UNDER CONTROL"

1.3B. CONFIGURE APPLE APNS FOR IOS MANAGEMENT

- **iOS and iPadOS require Apple Push Notification Service (APNs)**
  - Mandatory for all Apple device management
  - Required before any iOS enrollment can succeed
- **APNs establishes trust between Apple and Intune**
  - Allows Intune to send management commands
  - Apple controls the communication channel
- **APNs certificate is tenant-level**
  - One certificate per tenant
  - Affects all iOS/iPadOS devices
- **Certificate lifecycle must be managed**
  - Certificate expires annually
  - Expiration breaks device management
- **APNs must be active before enrolling iOS devices**
  - Enrollment fails without APNs
  - Existing devices stop responding if APNs expires

SECTION 2B — Mobile Device Enrollment (Android & iOS)

"BRING EVERY MOBILE DEVICE UNDER CONTROL"

1.4B. ENROLL IOS DEVICES USING COMPANY PORTAL

- **Enrollment is performed using Company Portal**
  - User signs in with corporate account
  - Guided enrollment experience
- **MDM management profile is installed**
  - Grants Intune management permissions
  - Required for device control
- **User explicitly approves device management**
  - Apple requires user consent
  - Transparency into what is managed
- **Device appears in Intune after enrollment**
  - Platform, ownership, and management state visible
  - Compliance status evaluated
- **Device becomes ready for enterprise security**
  - Policies, compliance, and Conditional Access
  - App management and protection enabled

SECTION 2B — Mobile Device Enrollment (Android & iOS)

"BRING EVERY MOBILE DEVICE UNDER CONTROL"

1.5B. VALIDATE AND COMPARE MOBILE DEVICE MANAGEMENT

- **Validate mobile devices in Intune after enrollment**
  - Confirm device appears in Intune
  - Enrollment must complete successfully
- **Review key device properties**
  - Platform and OS
  - Device ownership
  - Management type
- **Confirm compliance and device state**
  - Compliance evaluation status
  - Device health and management status
- **Compare mobile MDM vs Windows MDM**
  - Platform-specific enrollment requirements
  - Differences in control and visibility
- **Ensure devices are ready for security enforcement**
  - Policies, Conditional Access, and app protection
  - Prevent misconfigurations before moving forward

Lab 2.1B: Connect Managed Google Play (Android Enterprise)

STEP-BY-STEP INSTRUCTIONS

Lab 2.2B:
Enroll Android Device (Work Profile)

STEP-BY-STEP INSTRUCTIONS

Lab 2.3B: Configure Apple MDM Push Certificate (APNs)

STEP-BY-STEP INSTRUCTIONS

Lab 2.4B: Enroll iOS/iPadOS Device using Company Portal

STEP-BY-STEP INSTRUCTIONS

SECTION 3: Device Configuration Profiles

"TURN SECURITY REQUIREMENTS INTO REAL POLICIES ON REAL DEVICES."

SECTION 3 — Device Configuration Profiles

“TURN SECURITY REQUIREMENTS INTO REAL POLICIES ON REAL DEVICES.”

WHAT YOU WILL LEARN:-

- **3.1. Profile Types in Intune**
  - Understand what configuration profiles control on Windows
  - Choose between Settings Catalog, Templates, and Properties Catalog
  - Know when to configure settings vs when to collect device inventory
- **3.2. Assignments, Scope Tags, and Filters**
  - Target policies correctly using device and user assignments
  - Use scope tags to manage administrative visibility
  - Apply filters for dynamic, property-based targeting
- **3.3. Verification & Troubleshooting**
  - Validate policy deployment vs actual enforcement
  - Interpret policy status: success, pending, error, conflict
  - Troubleshoot using device behavior and MDM logs

SECTION 3 — Device Configuration Profiles

“TURN SECURITY REQUIREMENTS INTO REAL POLICIES ON REAL DEVICES.”

WHAT YOU WILL LEARN:-

- **Lab 3.1 — Create a Baseline Windows Hardening Profile (Settings Catalog)**
- **Lab 3.2 — Device Restrictions: Block USB & Control Features**
- **Lab 3.3 — Block Control Panel Using Administrative Templates**

SECTION 3 — Device Configuration Profiles

"TURN SECURITY REQUIREMENTS INTO REAL POLICIES ON REAL DEVICES."

3.1. PROFILE TYPES IN INTUNE

- **Why Configuration Profiles Matter**
  - Configuration profiles = enforced device behavior
  - Security settings, restrictions, system configuration
  - Foundation for all Intune policy work
- **Profile Types You'll Work With**
  - Settings catalog (modern + recommended)
  - Templates (structured + fast)
  - Properties catalog (inventory + visibility)
- **Settings Catalog (Recommended)**
  - Widest configuration coverage
  - Searchable and flexible
  - Best for modern Windows hardening
- **Templates (Pre-Grouped)**
  - Pre-grouped settings
  - Faster setup for common tasks
  - Ideal for restrictions and standard controls
- **Properties Catalog**
  - Collects device inventory
  - Hardware, OS, TPM, BIOS data
  - Does NOT enforce settings
- **How to Choose the Right Type**
  - Use Settings Catalog for configuration
  - Use Templates for ADMX-backed controls
  - Use Properties Catalog for insight, not enforcement

SECTION 3 — Device Configuration Profiles

"TURN SECURITY REQUIREMENTS INTO REAL POLICIES ON REAL DEVICES."

3.2. ASSIGNMENTS, SCOPE TAGS, AND FILTERS

Why Targeting Matters

- Assignments, Scope Tags, and Filters
  - Profiles don't work without correct targeting
  - Who gets the policy matters as much as the policy itself

Assignments

- Define who receives the policy
- Device groups vs user groups
- Device-based targeting is more predictable

Scope Tags

- Administrative boundaries
- Control visibility, not behavior
- Used in large environments

Filters

- Dynamic targeting
- Based on device properties
- Reduces complex group management

SECTION 3 — Device Configuration Profiles

“TURN SECURITY REQUIREMENTS INTO REAL POLICIES ON REAL DEVICES.”

3.3. VERIFICATION & TROUBLESHOOTING

Why Verification Matters

- Verification & Troubleshooting
- Deployment ≠ enforcement
- Always verify policy application
- Prevent silent failures

Policy Status in Intune

- Success
- Pending
- Error / Conflict

Endpoint Validation

- Device check-in status
- Local device behavior
- Event logs (MDM)

Timing & Sync Behavior

- Policies apply asynchronously
- Check-in schedules
- Manual sync for testing

LAB 3.1: Create a Baseline Windows Hardening Profile (Settings Catalog)

STEP-BY-STEP INSTRUCTIONS

LAB 3.2: Device Restrictions: Block USB & Control Features

STEP-BY-STEP INSTRUCTIONS

LAB 3.3: Block Control Panel using Administrative Templates

STEP-BY-STEP INSTRUCTIONS

SECTION 3 — Device Configuration Profiles

"TURN SECURITY REQUIREMENTS INTO REAL POLICIES ON REAL DEVICES."

LET'S WRAP-UP SECTION 3:-

- **Why Verification Matters**
  - Verification & Troubleshooting
  - Deployment ≠ enforcement
  - Always verify policy application
  - Prevent silent failures
- **Configuration profiles enforce how devices behave**
- **Three profile types serve different needs**
  - Settings catalog
  - Templates
  - Custom OMA-URI
- **Assignments, filters, and scope tags control targeting**
- **Policies apply asynchronously and must be verified**
- **Configuration profiles form the foundation of endpoint hardening**

SECTION 4: Compliance Policies & Conditional Access

"ENSURE ONLY TRUSTED USERS AND TRUSTED DEVICES ACCESS YOUR DATA."

SECTION 4— Compliance Policies & Conditional Access

"ENSURE ONLY TRUSTED USERS AND TRUSTED DEVICES ACCESS YOUR DATA."

WHAT YOU WILL LEARN:-

- **4.1. Device Compliance Fundamentals**
  - How Intune defines compliance and evaluates device health
- **4.2. Building Windows Compliance Policies**
  - Creating rules for encryption, firewall, antivirus, and OS health
- **4.3. Conditional Access + Compliance**
  - Using device compliance to allow or block access
- **4.4. Testing Compliant vs Non-Compliant Behavior**
  - Observing access decisions in real scenarios

SECTION 4— Compliance Policies & Conditional Access

“ENSURE ONLY TRUSTED USERS AND TRUSTED DEVICES ACCESS YOUR DATA.”

4.1. DEVICE COMPLIANCE FUNDAMENTALS

- **What Is Device Compliance in Intune?**
  - Compliance = health evaluation, not configuration
  - Continuous assessment of device security posture
- **What Does Intune Check for Compliance?**
  - Encryption status (BitLocker)
  - Firewall and antivirus state
  - Operating system version
  - Device health signals
- **Compliant vs Non-Compliant Devices**
  - Compliant = meets all requirements
  - Non-compliant = one or more checks fail
  - Compliance state updates automatically
- **What Happens When Compliance Fails**
  - Device marked as non-compliant
  - Enforcement actions triggered
  - Access can be restricted
- **Compliance and Zero Trust**
  - Never trust, always verify
  - Identity + device health together
  - Continuous enforcement

SECTION 4— Compliance Policies & Conditional Access

"ENSURE ONLY TRUSTED USERS AND TRUSTED DEVICES ACCESS YOUR DATA."

4.2. BUILDING WINDOWS COMPLIANCE POLICIES

What Is a Windows Compliance Policy?

- Defines what a "healthy" device looks like
- Evaluates device state, not configuration
- Determines whether a device is trusted

Common Compliance Requirements

- Encryption (BitLocker enabled)
- Antivirus and firewall protection
- Minimum operating system version

Compliance Actions & Grace Periods

- Immediate non-compliance vs grace period
- Time allowed for user remediation
- Balance between security and productivity

User Notifications & Remediation

- Notify users when devices are non-compliant
- Explain why access is restricted
- Guide users toward remediation

Compliance as an Input to Conditional Access

- Compliance alone does not block access
- Conditional Access enforces decisions
- Intune evaluates → Entra ID enforces

SECTION 4— Compliance Policies & Conditional Access

“ENSURE ONLY TRUSTED USERS AND TRUSTED DEVICES ACCESS YOUR DATA.”

4.3. CONDITIONAL ACCESS + DEVICE COMPLIANCE

- **Topic Introduction**
  - Identity alone is not enough
  - Device health influences access
  - Compliance feeds Conditional Access
- **What Conditional Access Actually Does**
  - Access decisions at sign-in time
  - Uses signals, not assumptions
  - Works across Microsoft cloud Apps
- **How Compliance Feeds Conditional Access**
  - Intune evaluates device health
  - Compliance result sent to Entra ID
  - Used during sing-in evaluation
- **Typical Compliance-Based CA Rule**
  - User signs in
  - Device Compliance Device
  - Access allowed or blocked
- **Why This Matters for Zero Trust**
  - Continuous verification
  - No permanent trust
  - Access changes dynamically

SECTION 4— Compliance Policies & Conditional Access

"ENSURE ONLY TRUSTED USERS AND TRUSTED DEVICES ACCESS YOUR DATA."

4.4. TESTING COMPLIANT VS NON-COMPLIANT BEHAVIOR

- **Topic Introduction**
  - Validate real enforcement
  - Observe user experience
  - Confirm Zero Trust behavior
- **Access When Device is Compliant**
  - Device meets requirements
  - Conditional Access allows access
  - No user disruption
- **What Happens When Compliance Break**
  - One requirement fails
  - Device marked non-compliant
  - Status updates automatically
- **Access Blocked in Real Time**
  - Identity is valid
  - Device health fails
  - Access denied
- **Restoring Compliance Restores Access**
  - Issue fixed
  - Device re-evaluated
  - Access allowed again
- **Why This Matters**
  - Confirms policy behavior
  - Prevents false confidence
  - Mirrors real incidents

LAB 4.1: Create a Windows Compliance Policy

STEP-BY-STEP INSTRUCTIONS

LAB 4.1:
Verify & Resolve “BitLocker Not Compliant”

PLACEMENT: IMMEDIATELY AFTER LAB 4.1

Lab 4.2: Create a Conditional Access Policy Using Device Compliance

STEP-BY-STEP INSTRUCTIONS

LAB 4.3: Test Compliant vs Non-Compliant Device Access

STEP-BY-STEP INSTRUCTIONS

SECTION 4— Compliance Policies & Conditional Access

“ENSURE ONLY TRUSTED USERS AND TRUSTED DEVICES ACCESS YOUR DATA.”

LET’S WRAP UP SECTION 4:-

- **Defined what *device compliance* means in Microsoft Intune**
- **Built a Windows compliance policy to evaluate device health**
- **Connected compliance status to Conditional Access**
- **Observed access allowed and blocked in real time**
- **Demonstrated Zero Trust enforcement lifecycle**

Section 5: Security Baselines

"APPLY MICROSOFT-RECOMMENDED SECURITY INSTANTLY—NO GUESSWORK."

SECTION 5 — What You Will Learn (Security Baselines)

"APPLY MICROSOFT-RECOMMENDED SECURITY INSTANTLY—NO GUESSWORK."

WHAT YOU WILL LEARN:-

- **5.1. Microsoft Security Baselines**
  - What security baselines are
  - Why Microsoft provides them
- **5.2. Baselines vs Configuration Profiles**
  - Prescriptive vs flexible security
  - When to use each approach
- **5.3. What Baselines Configure**
  - Defender and endpoint protection
  - Credential and system hardening
- **5.4. Applying a Security Baseline**
  - Where baselines live in Intune
  - Platform and version awareness
- **5.5. Baseline Conflicts & Overlap**
  - Why conflicts happen
  - How to detect and fix them

SECTION 5 — What You Will Learn (Security Baselines)

“APPLY MICROSOFT-RECOMMENDED SECURITY INSTANTLY—NO GUESSWORK.”

5.1. MICROSOFT SECURITY BASELINES

What Are Security Baselines?

- Microsoft-recommended security configurations
- Predefined collections of security settings
- Applied as a single policy
- Designed as a minimum secure posture

Why Microsoft Provides Baselines

- Built from real-world attack data
- Based on Microsoft threat intelligence
- Reflect enterprise security best practices
- Reduce guesswork for administrators

SECTION 5 — What You Will Learn (Security Baselines)

“APPLY MICROSOFT-RECOMMENDED SECURITY INSTANTLY—NO GUESSWORK.”

5.2. BASELINES VS CONFIGURATION PROFILES

- **Security Baselines**
  - Prescriptive and standardized
  - Microsoft-defined security decisions
  - Fast, consistent security hardening
- **Configuration Profiles**
  - Flexible and customizable
  - Admin-defined settings
  - Fine-grained control
- **When to Use Each**
  - Baselines → security foundation
  - Profiles → business-specific requirements

SECTION 5 — What You Will Learn (Security Baselines)

“APPLY MICROSOFT-RECOMMENDED SECURITY INSTANTLY—NO GUESSWORK.”

5.3. WHAT A SECURITY BASELINE ACTUALLY CONFIGURES

Understanding the real impact of baseline deployment

Baselines Are Broad, Not Small

- A baseline applies many security controls at once
- Affects multiple security domains simultaneously
- Can noticeably change device behavior
- Not just a “template”

Defender & Endpoint Protection

- Real-time protection enforcement
- Cloud-delivered protection enabled
- User ability to disable protection restricted
- Consistent Defender posture across devices

Credential Protection & Authentication

- Reduced credential exposure on endpoints
- Hardened authentication behavior
- Protection against credential theft and reuse
- Complements MFA (endpoint vs sign-in protection)

SECTION 5 — What You Will Learn (Security Baselines)

"APPLY MICROSOFT-RECOMMENDED SECURITY INSTANTLY—NO GUESSWORK."

5.4. APPLYING A SECURITY BASELINE

Where Security Baselines Live

- Separate from configuration profiles
- Dedicated Security Baselines section
- Platform-specific location

Platform & Version Awareness

- Windows 10 vs Windows 11 baselines
- Baselines are versioned
- New versions released over time

Review Before Deployment

- Settings are opinionated
- May impact apps or user experience
- Testing and review required

Assignment & Rollout Strategy

- Assigned like other Intune policies
- Pilot groups recommended
- Gradual rollout best practice

SECTION 5 — What You Will Learn (Security Baselines)

"APPLY MICROSOFT-RECOMMENDED SECURITY INSTANTLY—NO GUESSWORK."

5.5. BASELINE CONFLICTS & OVERLAP

Why Baseline Conflicts Happen

- Baselines configure many settings at once
- Other policies may manage the same settings
- Overlap with configuration profiles or endpoint security

How Conflicts Appear in Intune

- Policy conflicts reported in status views
- Unexpected device behavior
- Settings not applying as expected

How Intune Handles Conflicts

- Policy precedence rules
- Last-write or strongest-setting behavior
- Platform-specific resolution logic

Best Practices to Avoid Issues

- Review baseline settings before deployment
- Use pilot groups
- Avoid duplicating settings across policies

SECTION 5 — What You Will Learn (Security Baselines)

"APPLY MICROSOFT-RECOMMENDED SECURITY INSTANTLY—NO GUESSWORK."

5.3. WHAT A SECURITY BASELINE ACTUALLY CONFIGURES

Operating System & System Hardening

- OS-level security behaviors enforced
- System protections strengthened
- Reduced attack surface
- Potential impact on usability or legacy apps

Scale & Enterprise Impact

- Dozens of settings applied in one policy
- Faster than building multiple configuration profiles
- Consistent security at scale
- Requires careful review and testing

Key Takeaway

- Security baselines are powerful
- Immediate impact on device security posture
- Must be deployed deliberately
- Pilot testing is a best practice

SECTION 5 — What You Will Learn (Security Baselines)

“APPLY MICROSOFT-RECOMMENDED SECURITY INSTANTLY—NO GUESSWORK.”

5.4. MICROSOFT SECURITY BASELINES

What Are Microsoft Security Baselines?

- Microsoft-recommended security settings
- Predefined, standardized configurations
- Apply many security controls at once
- Designed as a minimum secure posture

Why Microsoft Provides Security Baselines

- Based on real-world attack data
- Built from Microsoft threat intelligence
- Designed for enterprise environments
- Reduce design guesswork for admins

What Baselines Are — and Are Not

- Baselines = starting point, not final design
- Opinionated by Microsoft
- Not tailored to every environment
- Intended to be reviewed before use

LAB 5.1: Apply a Microsoft Security Baseline to Windows 11

STEP-BY-STEP INSTRUCTIONS

LAB 5.2:
Analyze Baseline Conflicts & Remediate Errors

STEP-BY-STEP INSTRUCTIONS

SECTION 5 — What You Will Learn (Security Baselines)

“APPLY MICROSOFT-RECOMMENDED SECURITY INSTANTLY—NO GUESSWORK.”

LET’S WRAP-UP SECTION 5:-

- **Security baselines define a standardized security foundation**
- **Baselines are Microsoft-recommended and threat-driven**
- **Baselines differ from configuration profiles in purpose**
- **Baselines configure multiple security domains at once**
- **Careful deployment prevents conflicts and disruption**

Section 6: Microsoft Defender for Endpoint Integration

“TURN EVERY DEVICE INTO A MONITORED AND PROTECTED SECURITY ASSET.”

Section 6 — Microsoft Defender for Endpoint Integration

"TURN EVERY DEVICE INTO A MONITORED AND PROTECTED SECURITY ASSET."

WHAT YOU WILL LEARN IN SECTION 6:-

6.1. What Is Microsoft Defender for Endpoint

- What Defender for Endpoint actually is (EDR, not just antivirus)
- Difference between Defender Antivirus and Defender for Endpoint
- Why endpoint detection and response is critical in Zero Trust
- Defender's role beyond prevention

6.2. Defender for Endpoint Architecture & Integration

- Defender cloud architecture and telemetry flow
- How Defender integrates with Intune
- How device signals flow into Microsoft security services
- Why Defender integration is required for risk-based access decisions

6.3. Device Risk Levels & Threat Signals

- What device risk means in Defender
- Low, Medium, and High-risk levels
- What activities trigger device risk
- Difference between compliance status and device risk

6.4. Defender + Intune + Conditional Access

- How Defender risk influences Intune device state
- How device health feeds Conditional Access decisions
- Blocking access based on real-time threats
- Continuous enforcement under Zero Trust

Section 6 — Microsoft Defender for Endpoint Integration

“TURN EVERY DEVICE INTO A MONITORED AND PROTECTED SECURITY ASSET.”

WHAT YOU WILL LEARN IN SECTION 6:-

6.5. Onboarding Devices to Defender

- What onboarding to Defender means
- Intune-based onboarding vs other methods
- Required prerequisites for successful onboarding
- Common onboarding failures and how to avoid them

6.6. Threat Detection, Alerts & Automated Response

- Defender alerts and incidents
- Automated investigation and response (AIR)
- Device isolation and remediation actions
- How Defender responds without manual intervention

6.7. Defender Telemetry, Timeline & Validation

- Defender device timeline and attack story
- How to validate Defender is actively protecting a device
- What healthy Defender telemetry looks like
- How SOC teams confirm endpoint protection is working

Section 6 — Microsoft Defender for Endpoint Integration

“TURN EVERY DEVICE INTO A MONITORED AND PROTECTED SECURITY ASSET.”

6.1. WHAT IS MICROSOFT DEFENDER FOR ENDPOINT

- **What Is Microsoft Defender for Endpoint?**
  - Endpoint Detection & Response (EDR) platform
  - Not just antivirus
  - Continuous monitoring of device behavior
- **Defender Antivirus vs Defender for Endpoint**
  - Defender Antivirus = prevention
  - Defender for Endpoint = detection + response
  - Local protection vs cloud intelligence
- **Why EDR Is Critical in Zero Trust**
  - Modern attacks avoid traditional malware
  - Living-off-the-land techniques
  - Detection matters as much as prevention
- **Defender’s Role Beyond Prevention**
  - Behavioral detection
  - Attack correlation over time
  - Automated investigation and response
- **Continuous Monitoring & Risk Awareness**
  - Continuous evaluation
  - Not a one-time security check
  - Trust is always reassessed
- **What Defender Does (and Does Not Do)**
  - Provides threat and risk signals
  - Does NOT manage configuration
  - Feeds Intune and Entra ID decision

Section 6 — Microsoft Defender for Endpoint Integration

"TURN EVERY DEVICE INTO A MONITORED AND PROTECTED SECURITY ASSET."

6.2. DEFENDER FOR ENDPOINT ARCHITECTURE & INTUNE INTEGRATION

- **Defender for Endpoint Architecture Overview**
  - Cloud-based security architecture
  - Endpoint telemetry collection
  - Continuous analysis in Microsoft Security Cloud
- **Endpoint Telemetry & Signal Collection**
  - Process and memory activity
  - Network connections and behavior
  - File, registry, and system events
- **Microsoft Security Cloud Analysis**
  - Cloud analytics and threat intelligence
  - Behavioral and pattern-based detection
  - Correlation across devices and tenants
- **How Defender Integrates with Intune**
  - Intune manages configuration and onboarding
  - Defender provides threat and risk signals
  - Unified endpoint security model
- **Signal Flow into Microsoft Security Services**
  - Device → Defender → Microsoft Security services
  - Shared signals across Defender, Intune, Entra ID
  - Single security ecosystem
- **Why Defender Integration Enables Risk-Based Access**
  - Real-time device risk awareness
  - Dynamic trust decisions
  - Required for Zero Trust enforcement

Section 6 — Microsoft Defender for Endpoint Integration

“TURN EVERY DEVICE INTO A MONITORED AND PROTECTED SECURITY ASSET.”

6.3. DEVICE RISK LEVELS & THREAT SIGNALS

Device Risk in Microsoft Defender for Endpoint

- What device risk means in Defender
- Likelihood of compromise or active threat
- Based on observed behavior, not configuration

Defender Device Risk Levels

- Low risk
- Medium risk
- High risk
- Risk levels are dynamic

What Triggers Device Risk

- Malware execution
- Suspicious PowerShell or script activity
- Credential theft indicators
- Command-and-control communication
- Abnormal process behavior

Device Compliance vs Device Risk

- Compliance = configuration & health checks
- Risk = real-time threat activity
- Compliant device can still be risky
- Different signals, different purposes

Device Risk in Zero Trust

- Risk can override compliance
- High risk = reduced trust
- Used in Conditional Access decisions
- Continuous evaluation

Dynamic Risk & Remediation

- Risk increases when threats are detected
- Risk decreases after remediation
- Trust can be restored automatically
- Supports “never trust, always verify”

Section 6 — Microsoft Defender for Endpoint Integration

"TURN EVERY DEVICE INTO A MONITORED AND PROTECTED SECURITY ASSET."

6.4. DEFENDER + INTUNE + CONDITIONAL ACCESS

Roles in Zero Trust Enforcement

- Intune: device management & posture
- Defender for Endpoint: threat detection & device risk
- Conditional Access: access enforcement
- Each service has a distinct role

Separation of Detection and Enforcement

- Defender does NOT block access directly
- Defender generates device risk signals
- Conditional Access consumes those signals
- Clean separation of responsibilities

How Defender Risk Affects Device Trust

- Compliant ≠ trusted
- High Defender risk overrides compliance
- Configuration alone is not enough
- Trust is based on behavior

Conditional Access with Defender Risk

- Require compliant device
- Require low device risk
- Block Medium or High-risk devices
- Dynamic access decisions

Real-Time Enforcement

- Defender detects threat
- Device risk increases
- Access blocked automatically
- No manual intervention required

Continuous Evaluation in Zero Trust

- Trust is continuously reassessed
- Risk can increase or decrease
- Access restored after remediation
- Adaptive security model

Section 6 — Microsoft Defender for Endpoint Integration

"TURN EVERY DEVICE INTO A MONITORED AND PROTECTED SECURITY ASSET."

6.4. DEFENDER + INTUNE + CONDITIONAL ACCESS

- **Closed-Loop Security Model**
  - Defender: detect threats
  - Intune: manage device posture
  - Conditional Access: enforce decisions
  - Detect → Evaluate → Enforce → Recover

- **Topic 6.4 Summary**
  - Defender provides real-time risk
  - Intune provides device posture
  - Conditional Access enforces trust
  - Foundation of Zero Trust access control

Section 6 — Microsoft Defender for Endpoint Integration

“TURN EVERY DEVICE INTO A MONITORED AND PROTECTED SECURITY ASSET.”

6.5. ONBOARDING DEVICES TO DEFENDER

What Does “Onboarding” to Defender Mean?

- Enabling Defender sensors on the device
- Allowing telemetry to flow to Microsoft Security Cloud
- Required before detection, alerts, or response can occur

What Happens During Defender Onboarding

- Defender sensor activates on the device
- Telemetry begins streaming securely
- Device appears in Defender portal

Onboarding Methods Available

- Intune-based onboarding (recommended)
- Script-based onboarding
- Group Policy onboarding (legacy)

Intune-Based Defender Onboarding

- Uses endpoint security policies
- Centralized deployment
- Best for managed Windows devices

Required Prerequisites for Successful Onboarding

- Valid Defender for Endpoint license
- Device enrolled in Intune
- Supported Windows version
- Correct onboarding policy assigned

Section 6 — Microsoft Defender for Endpoint Integration

“TURN EVERY DEVICE INTO A MONITORED AND PROTECTED SECURITY ASSET.”

6.5. ONBOARDING DEVICES TO DEFENDER

- **Common Onboarding Failures**
  - Missing or incorrect license
  - Policy not assigned to device
  - Device not fully enrolled
  - Delayed policy sync
- **Verifying Successful Onboarding**
  - Device visible in Defender portal
  - Onboarding status = Success
  - Telemetry and timeline available
- **Topic 6.5 Summary**
  - Onboarding enables Defender protection
  - Intune is the recommended method
  - Prerequisites must be met
  - Verification is mandatory

Section 6 — Microsoft Defender for Endpoint Integration

"TURN EVERY DEVICE INTO A MONITORED AND PROTECTED SECURITY ASSET."

6.6. THREAT DETECTION, ALERTS & AUTOMATED RESPONSE

How Defender Detects Threats

- Continuous endpoint telemetry analysis
- Behavioral detection (not just signatures)
- Detection of modern, stealthy attack techniques

Defender Alerts

- Alerts represent specific security findings
- Include device, time, activity, and risk details
- Provide visibility into suspicious behavior

Incidents in Microsoft Defender

- Multiple alerts grouped into incidents
- Shows the full attack context
- Reduces alert noise for analysts

Automated Investigation & Response (AIR)

- Automated investigation of incidents
- Analysis of artifacts and system state
- Determines malicious vs benign activity

Automated Response Actions

- Stop malicious processes
- Remove or remediate files
- Isolate device from the network
- Contain threats quickly

Automation with Human Oversight

- Automation does not replace analysts
- Security teams can review and intervene
- Balance between speed and control

Section 6 — Microsoft Defender for Endpoint Integration

“TURN EVERY DEVICE INTO A MONITORED AND PROTECTED SECURITY ASSET.”

6.6. THREAT DETECTION, ALERTS & AUTOMATED RESPONSE

Speed and Impact of Automated Response

- Rapid response in minutes or seconds
- Prevents lateral movement and data loss
- Limits attack damage early

Visibility in the Defender Portal

- Full view of alerts and incidents
- Investigation and response history
- Data for tuning and improvement

Topic 6.6 Summary

- Behavioral threat detection
- Alerts grouped into incidents
- Automated investigation and response
- Fast containment with full visibility

Section 6 — Microsoft Defender for Endpoint Integration

“TURN EVERY DEVICE INTO A MONITORED AND PROTECTED SECURITY ASSET.”

6.7. TOPIC 6.7 — DEFENDER TELEMETRY, TIMELINE & VALIDATION

What Is Defender Telemetry?

- Security signals collected from endpoints
- Process, network, file, and system activity
- Foundation for detection and investigation

How Telemetry Is Used by Defender

- Feeds detection and risk evaluation
- Powers alerts, incidents, and timelines
- Continuously updated

The Defender Device Timeline

- Chronological view of device activity
- Processes, network events, alerts
- Visual attack story

Using the Timeline for Investigation

- Understand how an attack started
- Trace lateral or follow-up activity
- Validate automated response

Validating Defender Is Actively Protecting a Device

- Device visible in Defender portal
- Active timeline data present
- Recent telemetry timestamps

What Healthy Defender Telemetry Looks Like

- Continuous event flow
- No telemetry gaps
- Risk and alerts update dynamically

Section 6 — Microsoft Defender for Endpoint Integration

“TURN EVERY DEVICE INTO A MONITORED AND PROTECTED SECURITY ASSET.”

6.7. TOPIC 6.7 — DEFENDER TELEMETRY, TIMELINE & VALIDATION

- **How SOC Teams Use Telemetry & Timeline**
  - Confirm protection is active
  - Investigate incidents
  - Support audit and reporting
- **Topic 6.7 Summary**
  - Telemetry enables detection and response
  - Timeline shows the full attack story
  - Validation confirms Defender is working
  - Essential for SOC operation

LAB 6.1: Onboard Windows Device to Microsoft Defender for Endpoint

STEP-BY-STEP INSTRUCTIONS

LAB 6.2: Trigger Defender Alerts & Investigate Incidents

STEP-BY-STEP INSTRUCTIONS

LAB 6.3: Automated Response, Device Isolation & Remediation

STEP-BY-STEP INSTRUCTIONS

LAB 6.4: Defender Device Risk + Conditional Access Enforcement

STEP-BY-STEP INSTRUCTIONS

Section 6 — Microsoft Defender for Endpoint Integration

“TURN EVERY DEVICE INTO A MONITORED AND PROTECTED SECURITY ASSET.”

LET’S WRAP-UP SECTION 6:-

- **Defender for Endpoint is EDR, not just antivirus**
- **Defender, Intune, and Entra ID work as one ecosystem**
- **Device risk is different from compliance**
- **Real-time threats change access decisions**
- **Defender enables continuous Zero Trust enforcement**

SECTION 7: Application Management with Microsoft Intune

“DELIVER THE RIGHT APPS TO THE RIGHT DEVICES—SECURELY AND EFFICIENTLY.”

Section 7 — Application Management with Microsoft Intune

“DELIVER THE RIGHT APPS TO THE RIGHT DEVICES—SECURELY AND EFFICIENTLY.”

WHAT YOU WILL LEARN IN SECTION 7:-

- **7.1. Application Types in Intune**
  - What application types Intune supports
  - Difference between Win32, MSI, Microsoft Store, and LOB apps
  - When to use each app type in real environments
  - Common mistakes when choosing app types
- **7.2. Win32 Application Packaging & Deployment**
  - What Win32 apps are and why they are most commonly used
  - How Win32 app packaging works (IntuneWin format)
  - Installation, uninstall, and detection logic
  - Why Win32 apps are preferred for enterprise software
- **7.3. Microsoft Store App Deployment**
  - How Intune integrates with the Microsoft Store
  - Difference between Store apps and traditional installers
  - Deploying Store apps using Intune
  - When Store apps are a good or bad choice
- **7.4. App Assignments & User Experience**
  - Required vs Available app deployments
  - User-based vs device-based targeting for apps
  - How assignments affect what users see
  - App installation timing and background behavior

Section 7 — Application Management with Microsoft Intune

“DELIVER THE RIGHT APPS TO THE RIGHT DEVICES—SECURELY AND EFFICIENTLY.”

WHAT YOU WILL LEARN IN SECTION 7:-

- **7.5. App Protection Policies (MAM)**
  - What App Protection Policies are
  - Difference between MDM and MAM
  - Protecting corporate data inside apps
  - Supporting BYOD without managing the full device

- **7.6. Validating App Deployment & Troubleshooting**
  - How to verify app installation success
  - Understanding app status and error reporting
  - Common app deployment failures
  - Where to troubleshoot when apps don’t install

Section 7 — Application Management with Microsoft Intune

“DELIVER THE RIGHT APPS TO THE RIGHT DEVICES—SECURELY AND EFFICIENTLY.”

7.1. APPLICATION TYPES IN INTUNE

- **Why Application Types Matter**
  - Intune supports multiple application types
  - App type selection affects deployment success
  - Wrong app type = failed installs and poor user experience
- **Win32 Applications**
  - Traditional Windows desktop applications
  - Packaged using IntuneWin format
  - Most common app type in enterprises
- **MSI Applications**
  - Windows Installer–based applications
  - Limited compared to Win32 apps
  - Used mainly for simple installers
- **Microsoft Store Applications**
  - Apps delivered via Microsoft Store
  - Modern app deployment model
  - Easier deployment, limited control
- **Line-of-Business (LOB) Applications**
  - Internal or custom-built apps
  - Direct upload without packaging
  - Platform-specific limitations
- **Choosing the Right App Type**
  - Enterprise apps → Win32
  - Simple installers → MSI
  - Modern apps → Store
  - Internal tools → LOB

Section 7 — Application Management with Microsoft Intune

“DELIVER THE RIGHT APPS TO THE RIGHT DEVICES—SECURELY AND EFFICIENTLY.”

7.2. WIN32 APPLICATION PACKAGING & DEPLOYMENT

Why Win32 Apps Matter

- Most enterprise Windows apps are Win32
- Required for advanced install logic
- Offers the highest deployment reliability

What Makes an App a “Win32 App”

- Traditional EXE or MSI installers
- Packaged into IntuneWin format
- Installed silently by Intune agent

Win32 App Packaging Process

- Source installer files
- Microsoft Win32 Content Prep Tool
- Output: .intunewin package

Installation, Uninstall & Detection Logic

- Install command
- Uninstall command
- Detection rules determine success

Why Detection Rules Are Critical

- Intune does not “watch” installs
- Detection = proof of installation
- Incorrect detection causes failures

Win32 Deployment Behavior

- Background installation
- Retry and remediation logic
- Required vs Available behavior

Section 7 — Application Management with Microsoft Intune

"DELIVER THE RIGHT APPS TO THE RIGHT DEVICES—SECURELY AND EFFICIENTLY."

7.3. MICROSOFT STORE APP DEPLOYMENT

- **Why Microsoft Store Apps Exist**
  - Modern app delivery model
  - Trusted Microsoft-signed applications
  - Reduced packaging and maintenance effort
- **Types of Microsoft Store Apps in Intune**
  - Microsoft Store (New)
  - Microsoft Store legacy (deprecated)
  - Store apps vs Win32 apps
- **How Store App Deployment Works**
  - App sourced directly from Microsoft Store
  - Intune manages assignment and delivery
  - Installation handled by Windows
- **Benefits of Store App Deployment**
  - No packaging required
  - Automatic updates
  - Lower operational overhead
- **Limitations of Store Apps**
  - Limited customization and install control
  - Fewer enterprise-grade apps
  - Less flexible than Win32 deployments
- **When to Use Store Apps vs Win32**
  - Store apps for simple, standard software
  - Win32 apps for complex enterprise needs
  - Choosing the right tool matters

Section 7 — Application Management with Microsoft Intune

"DELIVER THE RIGHT APPS TO THE RIGHT DEVICES—SECURELY AND EFFICIENTLY."

7.4. APPLICATION ASSIGNMENTS & USER EXPERIENCE

- **Why Assignments Matter**
  - Assignment defines who gets the app
  - Assignment defines how the app appears
  - Direct impact on user experience
- **Required vs Available Applications**
  - Required apps install automatically
  - Available apps are user-initiated
  - Different use cases for each
- **User-Based vs Device-Based Assignments**
  - User-based targeting for apps
  - Device-based targeting scenarios
  - Understanding assignment behavior
- **Application Delivery Experience**
  - Company Portal experience
  - Install timing and notifications
  - Background vs foreground installs
- **App Dependency & Order**
  - App dependencies
  - Why order matters
  - Preventing failed installs
- **Designing a Good User Experience**
  - Minimize disruption
  - Predictable installs
  - Aligning security with productivity

Section 7 — Application Management with Microsoft Intune

"DELIVER THE RIGHT APPS TO THE RIGHT DEVICES—SECURELY AND EFFICIENTLY."

7.5. APP PROTECTION POLICIES (MAM)

What Are App Protection Policies (MAM)?

- Application-level data protection
- No device enrollment required
- Focused on corporate data, not the device

MDM vs MAM (Key Difference)

- MDM = device control
- MAM = app & data control
- Works on managed and unmanaged devices

What MAM Protects

- Corporate data inside apps
- Copy / paste restrictions
- Save-as and data leakage controls
- App-level encryption

Supported Apps & Platforms

- Microsoft apps (Outlook, Teams, OneDrive)
- iOS, Android, Windows
- App must be MAM-enabled

MAM in BYOD Scenarios

- Personal devices stay personal
- No visibility into device content
- Corporate data stays protected

Conditional Access + MAM

- Require app protection
- Allow access without device enrollment
- Balance security and user privacy

Section 7 — Application Management with Microsoft Intune

“DELIVER THE RIGHT APPS TO THE RIGHT DEVICES—SECURELY AND EFFICIENTLY.”

7.5. APP PROTECTION POLICIES (MAM)

- **Why MAM Is Critical in Modern Security**
  - Remote work reality
  - Zero Trust for applications
  - Data-centric security model

Section 7 — Application Management with Microsoft Intune

“DELIVER THE RIGHT APPS TO THE RIGHT DEVICES—SECURELY AND EFFICIENTLY.”

7.6. VALIDATING APP DEPLOYMENT & TROUBLESHOOTING

Why Validation & Troubleshooting Matter

- App deployment ≠ app success
- Validation confirms real user experience
- Troubleshooting is a core Intune admin skill
- Most issues are configuration-related

- *Instructor uses this slide for context-setting*

How Intune App Deployment Works

- Devices check in on a schedule
- Apps are evaluated, not instantly pushed
- Delays are normal and expected
- SCCM-style “instant installs” do not apply

- Aligns with explanation of asynchronous behavior

Validating in the Intune Portal

App monitoring shows:

- Succeeded
- Pending
- Failed
- Portal shows what happened
- Portal does NOT always show why

- Sets expectations for portal limitations

Section 7 — Application Management with Microsoft Intune

“DELIVER THE RIGHT APPS TO THE RIGHT DEVICES—SECURELY AND EFFICIENTLY.”

7.6. VALIDATING APP DEPLOYMENT & TROUBLESHOOTING

Validating on the Endpoint

Required apps:

- Installed automatically
- May install silently

Available apps:

- Appear in Company Portal
- User-initiated install
- Portal + device validation both required
- Reinforces “portal ≠ reality” concept

Common Causes of App Deployment Failure

- Wrong assignment type (Required vs Available)
- App assigned to wrong group
- Detection rule failures (Win32 apps)
- App appears installed but isn’t

- *High-value real-world slide*

Dependencies & Install Order

- Some apps require prerequisites
- Missing dependencies cause silent failures
- Win32 apps commonly affected
- Intune supports dependency sequencing

- *Enterprise-grade troubleshooting point*

Section 7 — Application Management with Microsoft Intune

"DELIVER THE RIGHT APPS TO THE RIGHT DEVICES—SECURELY AND EFFICIENTLY."

7.6. VALIDATING APP DEPLOYMENT & TROUBLESHOOTING

Logs: Where Real Troubleshooting Happens

- Intune Management Extension logs (Windows)shows:
  - App evaluation
  - Install command
  - Detection logic
  - Error codes
- Logs explain failures portal cannot

◆ *Critical admin skill slide*

Device Context & Timing Factors

- Device connectivity matters
- User sign-in state affects installs
- VPN, reboot, sleep state can delay installs
- Not all failures are real failures

◆ *Reduces false troubleshooting*

Troubleshooting Mindset (Best Practice)

- Check assignment first
- Confirm app type & targeting
- Validate detection logic
- Review logs before changing anything
- Most issues are misconfiguration

◆ *Strong admin discipline slide*

Key Takeaways

- Validation is as important as deployment
- Portal + endpoint validation required
- Logs tell the real story
- Calm, methodical troubleshooting wins

◆ Transition slide to labs

LAB 7.1:
Deploy a Win32 App

STEP-BY-STEP INSTRUCTIONS

LAB 7.2: Break a Win32 App & Troubleshoot

STEP-BY-STEP INSTRUCTIONS

LAB 7.3:
Deploy Microsoft Store App

STEP-BY-STEP INSTRUCTIONS

LAB 7.4: App Protection Policy (MAM) for Microsoft Edge

STEP-BY-STEP INSTRUCTIONS

LAB 7.4: App Protection Policy (MAM) on iOS (Microsoft Edge)

STEP-BY-STEP INSTRUCTIONS

Section 7 — Application Management with Microsoft Intune

"DELIVER THE RIGHT APPS TO THE RIGHT DEVICES—SECURELY AND EFFICIENTLY."

LET'S WRAP-UP SECTION 7

- **Application Management & Protection with Intune**
- **Deployed enterprise applications using Win32**
- **Understood how assignments shape user experience**
- **Troubleshot real Win32 deployment failures**
- **Protected corporate data using App Protection Policies (MAM)**
- **Balanced productivity and security for managed & unmanaged devices**

SECTION 8: Endpoint Security (ASR, Firewall, EPM)

"STRENGTHEN YOUR DEFENSES WITH ENTERPRISE-GRADE SECURITY CONTROLS."

Section 8 — Endpoint Security (ASR, Firewall, EPM)

“STRENGTHEN YOUR DEFENSES WITH ENTERPRISE-GRADE SECURITY CONTROLS.”

WHAT YOU WILL LEARN IN SECTION 8:-

- **8.1. Attack Surface Reduction (ASR)**
  - What attack surface reduction actually means
  - Why ASR rules exist and what threats they stop
  - How ASR reduces common attack techniques
- **8.2. Microsoft Defender Firewall Management**
  - Why host-based firewall still matters
  - How Intune manages Defender Firewall policies
  - Controlling inbound and outbound network traffic
- **8.3. Endpoint Privilege Management (EPM)**
  - Why local admin rights are dangerous
  - How EPM removes standing admin access
  - Just-in-time elevation with audit and control
- **8.4. Endpoint Security as a Zero Trust Control**
  - Prevention vs detection at the endpoint
  - How ASR, Firewall, and EPM work together
  - Reducing blast radius even after compromise
- **8.5. Validation & Troubleshooting**
  - How to confirm endpoint security policies apply
  - Where to check ASR, firewall, and EPM status
  - Common misconfigurations and how to fix them

Section 8 — Endpoint Security (ASR, Firewall, EPM)

"STRENGTHEN YOUR DEFENSES WITH ENTERPRISE-GRADE SECURITY CONTROLS."

8.1. ATTACK SURFACE REDUCTION

- **What Is Attack Surface Reduction?**
  - What "attack surface" means on a Windows device
  - Why reducing attack paths matters more than detection
  - ASR as a preventive security control
- **Why Microsoft Introduced ASR**
  - Modern attacks abuse built-in tools, not malware
  - Living-off-the-land techniques (LOLBins)
  - ASR blocks techniques, not files
- **What ASR Rules Actually Do**
  - Block risky behaviors, not applications
  - Prevent abuse of scripts, macros, and system tools
  - Enforce secure defaults at the OS level
- **Common Threats ASR Is Designed to Stop**
  - Malicious Office macros
  - Credential theft techniques
  - Script-based malware
  - Lateral movement behaviors
- **ASR in a Zero Trust Model**
  - Assume compromise is possible
  - Reduce blast radius before detection
  - Prevention + detection working together

Section 8 — Endpoint Security (ASR, Firewall, EPM)

“STRENGTHEN YOUR DEFENSES WITH ENTERPRISE-GRADE SECURITY CONTROLS.”

8.2. MICROSOFT DEFENDER FIREWALL MANAGEMENT

What Is Microsoft Defender Firewall?

- Built-in, host-based firewall in Windows
- Controls inbound and outbound network traffic
- Enforced per device, not per network

Why Host-Based Firewalls Matter

- Perimeter firewalls are no longer enough
- Devices move between networks
- Zero Trust assumes hostile networks

Firewall Profiles in Windows

- Domain profile
- Private profile
- Public profile
- Why profile context matters

Inbound vs Outbound Rules

- Inbound traffic control
- Outbound traffic control
- Why outbound blocking is often overlooked

Defender Firewall in Zero Trust

- Network access is never implicitly trusted
- Firewall enforces least privilege networking
- Works with ASR, Defender, and Conditional Access

Section 8 — Endpoint Security (ASR, Firewall, EPM)

"STRENGTHEN YOUR DEFENSES WITH ENTERPRISE-GRADE SECURITY CONTROLS."

8.3. ENDPOINT PRIVILEGE MANAGEMENT (EPM)

What Is Endpoint Privilege Management?

- Just-in-time local admin access
- Removes permanent local admin rights
- Least privilege enforcement on endpoints

Why Local Admin Is a Major Risk

- Most attacks require elevated privileges
- Malware abuses admin permissions
- Credential theft and persistence risks

How EPM Works in Intune

- Standard users by default
- Temporary elevation when required
- Elevation controlled by policy and approval

Elevation Rules & Controls

- App-based elevation rules
- User-based or device-based targeting
- Audit and justification tracking

EPM and Zero Trust Endpoint Security

- No standing trust
- Privilege granted only when needed
- Continuous control and visibility

Section 8 — Endpoint Security (ASR, Firewall, EPM)

“STRENGTHEN YOUR DEFENSES WITH ENTERPRISE-GRADE SECURITY CONTROLS.”

8.4. ENDPOINT SECURITY AS A ZERO TRUST CONTROL

Zero Trust and the Endpoint

- Zero Trust is not a product
- Endpoints are a primary attack surface
- Device trust must be continuously evaluated

Why Endpoint Security Is Central to Zero Trust

- Identities alone are not enough
- Compromised devices undermine MFA
- Device health influences access decisions

Endpoint Signals Used in Zero Trust

- Compliance state
- Defender device risk level
- Security configuration and posture
- Real-time threat signals

Continuous Enforcement, Not One-Time Trust

- Trust is never permanent
- Device state can change at any time
- Access adapts dynamically

Intune + Defender + Entra ID = Zero Trust

- Intune configures and evaluates
- Defender detects and responds
- Entra ID enforces access
- Unified security control loop

Section 8 — Endpoint Security (ASR, Firewall, EPM)

"STRENGTHEN YOUR DEFENSES WITH ENTERPRISE-GRADE SECURITY CONTROLS."

8.5. VALIDATION & TROUBLESHOOTING

Why Validation Matters

- Policies don't apply instantly
- Portal status ≠ actual device state
- Verification prevents false confidence

Where to Validate Endpoint Security

- Intune device status
- Endpoint security policy reports
- Microsoft Defender portal
- Windows endpoint behavior

Validating on the Endpoint

- Windows Security indicators
- Device sync and check-in behavior
- User-visible vs silent enforcement

Common Endpoint Security Failures

- Policy assigned but not applied
- Conflicting policies
- Unsupported OS or SKU
- Timing and sync delays

Troubleshooting Strategy

- Start with scope and assignment
- Confirm device state and signals
- Use logs and timelines
- Fix cause, not symptom

LAB 8.1: Create ASR Policy (Block Office Macros)

STEP-BY-STEP INSTRUCTIONS

LAB 8.2: Create Firewall Policy (Allow/Block App)

STEP-BY-STEP INSTRUCTIONS

LAB 8.3: Test ASR, Validate Defender Logs

STEP-BY-STEP INSTRUCTIONS

Section 8 — Endpoint Security (ASR, Firewall, EPM)

"STRENGTHEN YOUR DEFENSES WITH ENTERPRISE-GRADE SECURITY CONTROLS."

LET'S WRAP-UP SECTION 8

- **Endpoint Security in Intune**
  - Attack Surface Reduction (ASR)
  - Microsoft Defender Firewall
  - Endpoint Privilege Management (EPM)
- **How These Controls Work Together**
  - Reduce attack paths
  - Enforce least privilege
  - Block unauthorized behavior
  - Protect even after compromise
- **Endpoint Security + Zero Trust**
  - Continuous verification
  - Device behavior matters
  - Risk-based enforcement
  - No permanent trust
- **What You Can Do Now**
  - Design layered endpoint protection
  - Validate real enforcement
  - Troubleshoot conflicts confidently
  - Align security with user productivity

SECTION 9: Update Management (Windows Update for Business)

“KEEP EVERY DEVICE UP TO DATE, STABLE, AND COMPLIANT—AUTOMATICALLY.”

Section 9 — Update Management (Windows Update for Business)

“KEEP EVERY DEVICE UP TO DATE, STABLE, AND COMPLIANT—AUTOMATICALLY.”

WHAT YOU WILL LEARN IN SECTION 9

- **9.1. Why Update Management Matters**
  - Security risks of unpatched devices
  - Updates as a core Zero Trust control
  - Relationship between updates, compliance, and risk
- **9.2. Windows Update for Business (WUfB) Basics**
  - What Windows Update for Business is
  - How Intune controls Windows updates
  - Difference between WUfB and WSUS
- **9.3. Update Rings & Deployment Strategy**
  - Update rings and servicing channels
  - Feature updates vs quality updates
  - Deferral periods and deadlines
- **9.4. User Experience & Restart Control**
  - Active hours and restart behavior
  - Balancing enforcement vs productivity
  - Minimizing disruption
- **9.5. Monitoring, Compliance & Troubleshooting**
  - Update status and reporting
  - Common update failures
  - How to validate updates applied successfully

Section 9 — Update Management (Windows Update for Business)

"KEEP EVERY DEVICE UP TO DATE, STABLE, AND COMPLIANT—AUTOMATICALLY."

9.1. WHY UPDATE MANAGEMENT MATTERS

- **Unpatched Devices = Real Risk**
  - Unpatched vulnerabilities are a top attack vector
  - Attackers exploit known flaws, not just zero-days
  - Missing updates increase exposure window
- **Updates Are a Security Control**
  - Windows updates contain security fixes
  - Delayed updates = extended risk
  - Outdated devices cannot be trusted
- **Update Management in Zero Trust**
  - Zero Trust requires continuous verification
  - Device trust changes over time
  - Updates keep security posture current
- **Updates, Compliance & Access**
  - Update status affects device compliance
  - Non-compliant devices can lose access
  - Conditional Access enforces update health
- **Security vs User Experience**
  - Poor update handling frustrates users
  - No updates increases security risk
  - Balance is required
- **Modern Update Management**
  - Staged deployments
  - Controlled restarts
  - Monitoring update results
  - Cloud-based management with Intune

Section 9 — Update Management (Windows Update for Business)

“KEEP EVERY DEVICE UP TO DATE, STABLE, AND COMPLIANT—AUTOMATICALLY.”

9.1. WHY UPDATE MANAGEMENT MATTERS

- **Key Takeaway**
  - Update management is not optional
  - It is a core security control
  - Directly impacts trust and access

Section 9 — Update Management (Windows Update for Business)

“KEEP EVERY DEVICE UP TO DATE, STABLE, AND COMPLIANT—AUTOMATICALLY.”

9.2. WINDOWS UPDATE FOR BUSINESS (WUFB) BASICS

- **What Is Windows Update for Business (WUfB)?**
  - Cloud-based Windows update management service
  - Designed for modern, cloud-first environments
  - Devices pull updates directly from Microsoft
  - No on-prem update infrastructure required
- **What WUfB Is (and Is Not)**
  - Intune does NOT host update binaries
  - Microsoft hosts updates in the cloud
  - Intune controls update behavior using policies
  - Policy engine (Intune) + Delivery engine (Microsoft)
- **Policy-Based Update Management**
  - Updates managed using policies, not approvals
  - Define when and how updates install
  - No manual patch-by-patch approval
  - Rules-based update control
- **Update Categories in WUfB**
  - Quality updates (monthly security fixes)
  - Feature updates (Windows version upgrades)
  - Driver updates
  - Optional updates
  - Each category managed independently
- **Staged Deployment with Update Rings**
  - Devices grouped into rollout rings
  - Early rings receive updates first
  - Later rings follow after validation
  - Reduces risk and large-scale disruption

Section 9 — Update Management (Windows Update for Business)

"KEEP EVERY DEVICE UP TO DATE, STABLE, AND COMPLIANT—AUTOMATICALLY."

9.2. WINDOWS UPDATE FOR BUSINESS (WUFB) BASICS

- **Restart & User Experience Control**
  - Active hours configuration
  - Restart deadlines and grace periods
  - User notifications and prompts
  - Balance security with productivity
- **WUfB in a Zero Trust Model**
  - Update posture affects device trust
  - Outdated devices increase risk
  - Update status feeds compliance and access decisions
  - Continuous verification through updates
- **Key Takeaways**
  - WUfB is Microsoft's modern update platform
  - Intune controls behavior, Microsoft delivers updates
  - Cloud-native, scalable, and remote-friendly
  - Foundation for secure, Zero Trust-aligned update management

Section 9 — Update Management (Windows Update for Business)

"KEEP EVERY DEVICE UP TO DATE, STABLE, AND COMPLIANT—AUTOMATICALLY."

9.3. UPDATE RINGS & DEPLOYMENT STRATEGY

What Is an Update Ring?

- Logical grouping of devices for update rollout
- Controls when devices receive updates
- Used to stage updates safely
- Core concept in Windows Update for Business

Why Update Rings Matter

- Reduce blast radius of bad updates
- Catch issues early with pilot devices
- Prevent organization-wide disruption
- Enable controlled, predictable updates

Common Update Ring Model

- Pilot / Test ring
- Broad / Production ring
- Optional Critical / Fast ring
- Each ring has different timing and rules

Key Ring Configuration Settings

- Update deferral periods
- Deadline and grace period
- Restart behavior and notifications
- User experience controls

Section 9 — Update Management (Windows Update for Business)

“KEEP EVERY DEVICE UP TO DATE, STABLE, AND COMPLIANT—AUTOMATICALLY.”

9.3. UPDATE RINGS & DEPLOYMENT STRATEGY

Deployment Strategy Best Practices

- Start small, expand gradually
- Separate IT and business users
- Monitor results before moving forward
- Never deploy to all devices at once

Update Rings in Zero Trust

- Update state affects device trust
- Out-of-date devices increase risk
- Rings support continuous verification
- Security + stability together

Key Takeaways

- Update rings control rollout timing
- Strategy matters more than speed
- Proper rings reduce risk and downtime
- Foundation for secure update management

Section 9 — Update Management (Windows Update for Business)

"KEEP EVERY DEVICE UP TO DATE, STABLE, AND COMPLIANT—AUTOMATICALLY."

9.4. USER EXPERIENCE & RESTART CONTROL

Why User Experience Matters in Update Management

- Updates impact productivity
- Poor restart handling causes frustration
- User experience affects update adoption

The Restart Problem

- Updates often require reboots
- Forced restarts interrupt work
- Users delaying restarts increases risk

Active Hours

- Define when restarts are not allowed
- Protect business hours
- Device-aware restart control

Restart Deadlines

- Maximum time before restart is enforced
- Balances security and flexibility
- Prevents indefinite postponement

User Notifications & Warnings

- Restart prompts and countdowns
- Clear communication to users
- Predictable restart behavior

Balancing Security & Productivity

- Too strict = disruption
- Too relaxed = risk
- Controlled, predictable enforcement

Section 9 — Update Management (Windows Update for Business)

“KEEP EVERY DEVICE UP TO DATE, STABLE, AND COMPLIANT—AUTOMATICALLY.”

9.5. MONITORING, COMPLIANCE & TROUBLESHOOTING

- **Why Monitoring Updates Matters**
  - Visibility into update status
  - Detect failures and delays
  - Validate security posture
- **Update Monitoring in Intune**
  - Update ring reports
  - Device update status
  - Success, pending, failed states
- **Update Compliance & Device Trust**
  - Update status affects compliance
  - Outdated devices increase risk
  - Compliance used in access decisions
- **Common Update Issues**
  - Devices stuck pending updates
  - Restart not completed
  - Network or power conditions
  - User deferrals
- **Troubleshooting Tools**
  - Intune device status views
  - Windows Update logs
  - Manual sync and restart checks
- **Best Practices for Troubleshooting**
  - Start with portal visibility
  - Validate on the endpoint
  - Don’t assume policy = applied

Section 9 — Update Management (Windows Update for Business)

“KEEP EVERY DEVICE UP TO DATE, STABLE, AND COMPLIANT—AUTOMATICALLY.”

9.5. MONITORING, COMPLIANCE & TROUBLESHOOTING

- **Updates in a Zero Trust Model**
  - Continuous verification
  - Update health feeds trust
  - Monitoring closes the loop

LAB 9.1: Create a Windows Update Ring

STEP-BY-STEP INSTRUCTIONS

LAB 9.2:
Create Feature Update Policy

STEP-BY-STEP INSTRUCTIONS

Section 9 — Update Management (Windows Update for Business)

“KEEP EVERY DEVICE UP TO DATE, STABLE, AND COMPLIANT—AUTOMATICALLY.”

LET’S WRAP-UP SECTION 9

- **Updates are a security control, not maintenance**
- **Windows Update for Business enables cloud-first update management**
- **Update rings reduce risk through staged deployment**
- **User experience and restart control matter**
- **Update posture impacts compliance and access**

SECTION 10: Monitoring, Reporting & Troubleshooting

"SEE EVERYTHING, FIX ANYTHING, AND STAY AHEAD OF ISSUES."

Section 10 — Monitoring, Reporting & Troubleshooting

"SEE EVERYTHING, FIX ANYTHING, AND STAY AHEAD OF ISSUES."

WHAT YOU WILL LEARN IN SECTION 10

- **10.1. Intune Monitoring & Troubleshooting Portal**
  - Use the Troubleshooting + support blade
  - Track user/device issues quickly
  - Identify policy assignment + last check-in
- **10.2. Reading Policy Deployment & Device Health Signals**
  - Profile status: Succeeded / Pending / Failed
  - What "Not applicable" actually means
  - Sync behavior and check-in timing
- **10.3. Proactive Remediation & Scripts**
  - Use remediation scripts to fix common issues
  - Detect vs remediate logic
  - Validate results on endpoint
- **10.4. Reporting & Exporting Evidence**
  - Export device compliance and configuration reports
  - Evidence for audits and management updates
  - What to include in a "SOC/IT-ready" report
- **10.5. Common Intune Problems & Real Fixes**
  - Enrollment issues and missing MDM authority
  - Policy conflicts and setting overlap
  - Why "device exists in Entra but not Intune" happens

Section 10 — Monitoring, Reporting & Troubleshooting

"SEE EVERYTHING, FIX ANYTHING, AND STAY AHEAD OF ISSUES."

10.1. INTUNE MONITORING & TROUBLESHOOTING PORTAL

Why the Troubleshooting Portal Matters

- Central place for real-world issue investigation
- Focused on users and devices
- Designed for fast problem isolation

What the Troubleshooting Portal Is

- User-centric troubleshooting view
- Correlates devices, policies, and assignments
- Real-time visibility into Intune signals

Key Information You Can See

- Assigned policies and profiles
- Device compliance and configuration status
- Last device check-in time
- Errors and deployment states

Common Scenarios It Solves

- "Policy not applying" complaints
- App not installing for a user
- Device out of compliance
- Enrollment and sync issues

What the Portal Does NOT Do

- Does not fix issues automatically
- Does not replace endpoint validation
- Does not override policy logic

Section 10 — Monitoring, Reporting & Troubleshooting

"SEE EVERYTHING, FIX ANYTHING, AND STAY AHEAD OF ISSUES."

10.2. READING POLICY DEPLOYMENT & DEVICE HEALTH SIGNALS

Why Reading Signals Correctly Matters

- Policies don't apply instantly
- Status values are often misunderstood
- Misreading signals leads to false troubleshooting

Core Policy Deployment States

- Succeeded
- Pending
- Failed
- Not applicable

What "Pending" Really Means

- Device hasn't checked in yet
- Policy queued but not processed
- Restart or sync may be required

Understanding "Failed" vs "Not Applicable"

- Failed = policy processed but errored
- Not applicable = policy doesn't apply to this device
- Platform, SKU, or setting mismatch

Device Health Signals in Intune

- Last check-in time
- Compliance state
- Management authority
- Primary user assignment

Timing, Sync, and Asynchronous Behavior

- Intune is not real-time
- Check-in cycles matter
- Manual sync for validation

Section 10 — Monitoring, Reporting & Troubleshooting

“SEE EVERYTHING, FIX ANYTHING, AND STAY AHEAD OF ISSUES.”

10.2. READING POLICY DEPLOYMENT & DEVICE HEALTH SIGNALS

- **Signal Interpretation in Zero Trust**
  - Trust is dynamic
  - Health signals feed access decisions
  - Verification beats assumptions

Section 10 — Monitoring, Reporting & Troubleshooting

"SEE EVERYTHING, FIX ANYTHING, AND STAY AHEAD OF ISSUES."

10.3. PROACTIVE REMEDIATION & SCRIPTS

- **Why Proactive Remediation Exists**
  - Reactive troubleshooting doesn't scale
  - Devices drift over time
  - Fixing issues before users notice
- **What Proactive Remediation Is**
  - Detection script + remediation script
  - Runs on a schedule
  - Fixes issues automatically
- **Detection Scripts**
  - Identify unhealthy states
  - Return compliant / non-compliant result
  - PowerShell-based logic
- **Remediation Scripts**
  - Triggered only when detection fails
  - Correct misconfigurations
  - Restore desired state
- **Common Real-World Use Cases**
  - Restarting stuck services
  - Re-enabling Defender features
  - Fixing registry or configuration drift
- **Where Proactive Remediation Fits**
  - Complements policies
  - Not a replacement for configuration profiles
  - Operational reliability layer

Section 10 — Monitoring, Reporting & Troubleshooting

“SEE EVERYTHING, FIX ANYTHING, AND STAY AHEAD OF ISSUES.”

10.3. PROACTIVE REMEDIATION & SCRIPTS

- **Best Practices & Cautions**
  - Test scripts carefully
  - Avoid destructive actions
  - Monitor execution results

Section 10 — Monitoring, Reporting & Troubleshooting

"SEE EVERYTHING, FIX ANYTHING, AND STAY AHEAD OF ISSUES."

10.4. REPORTING & EXPORTING EVIDENCE

- **Why Reporting Matters**
  - Visibility into device health
  - Proof of security enforcement
  - Required for audits & compliance
- **Intune Reporting Capabilities**
  - Policy deployment reports
  - Device compliance reports
  - Application & update reports
- **Where to Find Key Reports**
  - Devices → Monitor
  - Reports → Endpoint analytics
  - Policy-specific status views
- **Exporting Evidence**
  - CSV and downloadable reports
  - Evidence for audits and management
  - Sharing with security & compliance teams
- **Common Evidence Use Cases**
  - Proving policy enforcement
  - Showing compliance over time
  - Supporting investigations
- **Limitations of Intune Reporting**
  - Near real-time, not instant
  - Retention limits
  - Context sometimes required

Section 10 — Monitoring, Reporting & Troubleshooting

“SEE EVERYTHING, FIX ANYTHING, AND STAY AHEAD OF ISSUES.”

10.4. REPORTING & EXPORTING EVIDENCE

- **Best Practices for Evidence Collection**
  - Combine multiple reports
  - Capture timestamps & scope
  - Keep exports organized

Section 10 — Monitoring, Reporting & Troubleshooting

“SEE EVERYTHING, FIX ANYTHING, AND STAY AHEAD OF ISSUES.”

10.5. COMMON INTUNE PROBLEMS & REAL FIXES

- **Why Intune Issues Happen**
  - Policy logic vs device reality
  - Timing, scope, and prerequisites
  - Misunderstanding “success”
- **Problem 1: Policy Not Applying**
  - Assigned but not enforced
  - Device shows “Pending” or “Not applicable”
  - Common root causes
- **Problem 2: Device in Entra ID but Not in Intune**
  - Enrollment misconceptions
  - MDM authority & auto-enrollment
  - Licensing and scope issues
- **Problem 3: Conflicting Policies**
  - Baselines vs profiles
  - Duplicate settings
  - Unexpected device behavior
- **Problem 4: Apps Fail or Reinstall Repeatedly**
  - Detection rule failures
  - Assignment mistakes
  - Dependency issues
- **Problem 5: Compliance & Access Issues**
  - Compliant but blocked
  - Non-compliant but allowed
  - Timing and signal mismatch

Section 10 — Monitoring, Reporting & Troubleshooting

“SEE EVERYTHING, FIX ANYTHING, AND STAY AHEAD OF ISSUES.”

10.5. COMMON INTUNE PROBLEMS & REAL FIXES

- **How to Fix Intune Issues Systematically**
  - Validate scope first
  - Check device signals
  - Trust evidence, not assumptions

Lab 10.1: Use Intune Troubleshooting Portal

STEP-BY-STEP INSTRUCTIONS

Lab 10.2: Deploy Custom Remediation Script

STEP-BY-STEP INSTRUCTIONS

Lab 10.2:
Export an Intune Management Report

STEP-BY-STEP INSTRUCTIONS

Section 10 — Monitoring, Reporting & Troubleshooting

"SEE EVERYTHING, FIX ANYTHING, AND STAY AHEAD OF ISSUES."

LET'S WRAP-UP SECTION 10:-

What You Mastered in Section 10

- Intune monitoring & troubleshooting portal
- Reading policy deployment and device health signals
- Using logs instead of assumptions
- Proactive remediation and scripts
- Identifying common Intune problems and real fixes

Key Takeaways

- Assignment ≠ enforcement
- Evidence > assumptions
- Logs tell the real story
- Troubleshooting is a repeatable process

Why This Section Matters

- Turns configuration into confidence
- Enables faster incident resolution
- Reduces trial-and-error changes
- Builds real-world Intune expertise